

- ①. Air is confined to one side of a rigid container divided by a partition, as shown in figure. The other side is initially evacuated. The air is initially at $p_1 = 5$ bar, $T_1 = 500$ K, and $V_1 = 0.2$ m³. When the partition is removed, the air expands to fill the entire chamber. Measurements show that $V_2 = 2V_1$ and $p_2 = p_1/4$. Assuming the air behaves as an ideal gas, determine (a) the final temperature, in K, and (b) the heat transfer, kJ.

